

Active Directory replication: what really happens

How a change on one domain controller reaches every other one — the topology the KCC builds, the notify-then-pull flow, and the ports it all depends on.

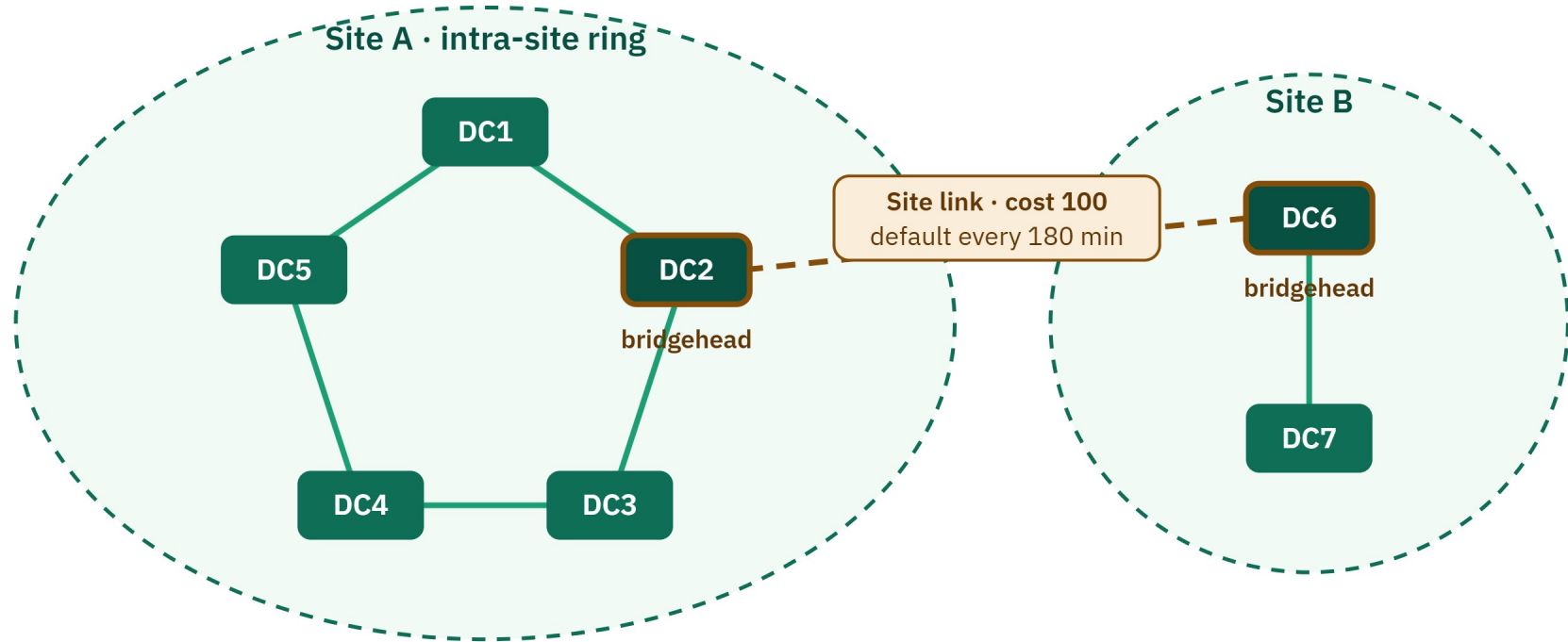
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1 What AD replication is

The process that keeps every domain controller's copy of the directory in sync, so a change made on one DC — a new user, a password reset, a group change — reaches the others.

- ✓ **Multi-master** — any writable DC accepts changes. Read-only DCs (RODCs) only receive.
- ✓ **Asynchronous** — DCs converge over time (loose consistency), not instantly.
- ✓ **Pull-based** — a DC announces a change; its partners request it over RPC.
- ✓ **Per partition** — each partition has its own topology and its own set of DCs that hold it.
- ✓ **Urgent replication** — security-sensitive changes such as account lockouts notify partners immediately within a site.

2 Topology: a ring inside, a link between



Within a site the KCC builds a bidirectional ring; at least two partners per DC, never more than three hops apart, with shortcut links added as the ring grows.

Between sites one DC per site (the ISTG) builds a spanning tree over site links; bridgehead servers carry the traffic.

3 What replicates, and to whom

PARTITION	HOLDS	REPLICATES TO
Domain	Users, computers, groups, OUs, Group Policy containers	DCs in that domain; GCs keep a partial copy
Configuration	Sites, site links, subnets, services	Every DC in the forest
Schema	Object classes and attributes	Every DC in the forest
Application	e.g. DomainDnsZones, ForestDnsZones	Only the DCs you enlist
SYSVOL	Logon scripts and GPO files (the GPT)	Every DC in the domain, via DFSR — not AD replication

4 The moving parts

NTDS.dit — the directory database on each DC.

USN — a per-DC counter that rises with every local write; partners use it to ask for "everything since."

KCC — runs on every DC (every 15 minutes by default) and builds the intra-site ring.

ISTG — one DC per site that builds the inter-site connections.

Connection objects — one-way, inbound: each says "this DC pulls from that one."

Sites, subnets, site links — map the physical network so the KCC knows what's fast and what's costly.

5 Intra-site vs inter-site

	INTRA-SITE	INTER-SITE
Topology	Bidirectional ring, ≤3 hops	Spanning tree over site links
Trigger	Change notification: first partner after 15 s, the rest 3 s apart	Schedule, default every 180 minutes
Built by	KCC on each DC	ISTG, via bridgeheads
Compression	None	Compressed by default
Transport	RPC over IP	RPC over IP

6 One change, step by step

01

Write on DC1

The change commits to NTDS.dit. DC1's USN rises and the attribute is stamped with a version, the originating DC and a timestamp.

02

Notify partners

In-site, DC1 waits 15 seconds, notifies its first partner, then the rest 3 seconds apart. Across sites, nothing moves until the schedule opens.

03

Partner pulls

DC2 requests changes since the last USN it saw from DC1, over RPC. Its up-to-dateness vector skips anything it already has.

04

Apply and pass on

DC2 writes the change under its own USN but keeps the original stamp, so conflicts resolve the same way everywhere. Then DC2 notifies its partners.

Conflicts: the higher version wins; on a tie, the later timestamp. Password changes are also forwarded straight to the PDC emulator.

7 Ports between DCs

PORT	SERVICE	USED FOR
53	TCP/UDP	DNS
88	TCP/UDP	Kerberos
123	UDP	W32Time
135	TCP	RPC EPM
389	TCP/UDP	LDAP
445	TCP	SMB
464	TCP/UDP	Kpasswd
636	TCP	LDAPS
3268 / 3269	TCP	Global catalog
49152–65535	TCP	Dynamic RPC

Common myths

Seen on popular diagrams

"The KCC builds a full mesh"

It's a bidirectional ring with shortcuts — at most three hops.

"DC1 pushes the change"

DC1 only notifies. Partners pull the data.

"DFSR uses SMB 445"

DFSR replicates SYSVOL over RPC. Clients read SYSVOL over SMB.

"Every DC is writable"

RODCs hold a read-only copy and never replicate outbound.